

# Pattern A

## Immunization Write Path — Tracer Bullet

End-to-end FHIR (CH VACD Immunization) → openEHR Composition → EHRbase, with feeder\_audit retention.

- PBLL (Kotlin + Ktor) — single brain orchestrating the stack
- fhir-server-1 (HAPI 8.8.1) — demographics + directory store (Patient / Practitioner / Organization)
- openFHIR 2.2.1 — stateless FHIR ↔ openEHR mapping engine (FHIRconnect YAMLS)
- EHRbase 2.31.0 — openEHR Composition store + AQL

Scope: 7 mapped Immunization fields (Tier 1 tracer bullet)

Out of scope: read path, profile validation, terminology server, SMART/EPR auth.

*Built as a Pragmatic Programmer 'tracer bullet': minimal slice, real infrastructure end-to-end.*

*Hackathon harness — see [services/pbll/](#) and [docs/architecture/patterns.md](#)*

# Stack at a Glance

Five services on a shared vacd-net bridge; PBLL is the only participant-built component.

| Service               | Role                                                            | URL inside vacd-net                                         | Auth                          |
|-----------------------|-----------------------------------------------------------------|-------------------------------------------------------------|-------------------------------|
| <b>PBLL</b>           | Platform brain (Kotlin/Ktor)                                    | http://pbll:8080                                            | —                             |
| <b>fhir-server-1</b>  | HAPI 8.8.1 — CH VACD ref. server, demographics + directory      | http://fhir-server-1:9111/ch-vacd-api-reference-server/fhir | —                             |
| <b>openfhir</b>       | openFHIR 2.2.1 — FHIR ↔ openEHR mapping (FHIRconnect)           | http://openfhir:8083                                        | —                             |
| <b>openfhir-mongo</b> | Mongo 7.0 — openFHIR config store (templates, contexts, models) | (internal)                                                  | admin/admin (internal)        |
| <b>ehrbase</b>        | EHRbase 2.31.0 — openEHR Composition store, AQL                 | http://ehrbase:8080/ehrbase/rest/openehr/v1                 | ehrbase-user / SuperSecret... |
| <b>ehrbase-db</b>     | Postgres 16 — EHRbase backing store                             | (internal)                                                  | internal                      |

# The Write Flow

POST /Immunization → fan out → ensure EHR → openFHIR convert → enrich → store

## 1. Client

POSTs a CH VACD Immunization (or document Bundle)

*to PBLL /Immunization*

## 2. Demographics fan-out

Patient + Practitioner + Organization → fhir-server-1 (HAPI)

*PBLL captures the HAPI-assigned ids*

## 3. EHR resolution

Find-or-create openEHR EHR in EHRbase

*Linked via EHR\_STATUS.subject.external\_ref (namespace=ch-vacd, id=<Patient.id>)*

## 4. Conversion

openFHIR /openfhir/toopenehr?templateId=...&flat=true

*Returns FLAT openEHR Composition shaped by ch-vacd-immunization administration.v1-alpha*

## 5. Feeder audit

PBLL injects /\_feeder\_audit/original\_content = verbatim FHIR JSON

*Per Konkretisierung §13 — legal provenance*

## 6. Persist

POST /ehr/{ehrId}/composition?format=FLAT&templateId=...

*Returns 201 + compositionUid; PBLL relays as 201 Created*

# Seven Mapped Fields (Tier 1)

Deliberately the alpha-safe subset — avoids every known alpha-quality issue in the template.

| # | FHIR path                                             | openEHR path                                                   | Type             |
|---|-------------------------------------------------------|----------------------------------------------------------------|------------------|
| 1 | Immunization.status = 'completed'                     | ism_transition/{careflow_step=at0006, current_state=245}       | manual           |
| 2 | Immunization.vaccineCode                              | description[at0017]/items[at0020]                              | CODEABLE_CONCEPT |
| 3 | Immunization.occurrenceDateTime                       | time                                                           | DATETIME         |
| 4 | Immunization.lotNumber                                | description[at0017]/items[CLUSTER.medication.v2]/items[at0150] | STRING           |
| 5 | Immunization.site                                     | description[at0017]/items[at0140]/items[at0141]                | CODEABLE_CONCEPT |
| 6 | Immunization.route                                    | description[at0017]/items[at0140]/items[at0147]                | CODEABLE_CONCEPT |
| 7 | Immunization.protocolApplied[0].doseNumberPositiveInt | description[at0017]/items[at0025]                              | INTEGER          |

# Demo UI — initial state

/demo loads with the Boostrix example pre-populated.

● CH VACD

Pattern A — Platform Business Logic Layer (tracer bullet)

healthz metadata CH VACD IG ↗

Example: Immunization administration — boostrix

Convert & Store ▶

1 · FHIR Immunization (input)

application/fhir+json

```
{
  "resourceType": "Bundle",
  "id": "1-1-ImmunizationAdministration",
  "meta": {
    "lastUpdated": "2021-06-01T00:00:00.374+02:00",
    "profile": ["http://fhir.ch/ig/ch-vacd/StructureDefinition/ch-vacd-document-immunization-administration"]
  },
  "identifier": {
    "system": "urn:ietf:rfc:3986",
    "value": "urn:uuid:5bca53b2-0973-4b91-961e-b1e93474c9c2"
  },
  "type": "document",
  "timestamp": "2021-06-01T00:00:00.390+02:00",
  "entry": [{
    "fullUrl":
"http://test.fhir.ch/r4/Composition/2-1-ImmunizationAdministrationComposition",
    "resource": {
      "resourceType": "Composition",
      "id": "2-1-ImmunizationAdministrationComposition",
      "meta": {
```

2 · openEHR FLAT Composition (after openFHIR)

intermediate

FLAT JSON will appear here.

3 · EHRbase confirmation

AQL roundtrip

Composition UID, EHR ID, AQL result.

How this works

1. Client POSTs a CH VACD Immunization (or document Bundle) to `POST /Immunization`.
2. PBLL fans Patient/Practitioner/Organization to `fhir-server-1` (HAPI). The FHIR server is the demographics + directory store.
3. PBLL ensures an openEHR EHR exists in `EHRbase`, linked to the Patient via `EHR_STATUS.subject.external_ref`.
4. PBLL calls `openFHIR /openfhir/toopenehr` with the Immunization to get a FLAT openEHR Composition.
5. PBLL enriches FLAT with `ctx/feeder_audit/original_content` (verbatim FHIR JSON — Konkretisierung §13).
6. PBLL persists the Composition to EHRbase and returns `201 Created` with `compositionUid`.

Hackathon harness — services: HAPI 8.8.1 · openFHIR 2.2.1 · EHRbase 2.31.0

screenshot: 00-initial.png

# Boostrix (dTpa-IPV booster) — CH VACD Bundle/document

The verbatim IG seed (fhir.ch/ig/ch-vacd/Bundle-1-1-ImmunizationAdministration); deltoid IM.

● CH VACD Pattern A — Platform Business Logic Layer (tracer bullet)

healthz metadata CH VACD IG ↗

Example: Immunization administration — boostrix ▼

Convert &amp; Store ▶

## 1 · FHIR Immunization (input)

application/fhir+json

```
{
  "resourceType": "Bundle",
  "id": "1-1-ImmunizationAdministration",
  "meta": {
    "lastUpdated": "2021-06-01T00:00:00.374+02:00",
    "profile": ["http://fhir.ch/ig/ch-vacd/StructureDefinition/ch-vacd-document-immunization-administration"]
  },
  "identifier": {
    "system": "urn:ietf:rfc:3986",
    "value": "urn:uuid:5bca53b2-0973-4b91-961e-b1e93474c9c2"
  },
  "type": "document",
  "timestamp": "2021-06-01T00:00:00.390+02:00",
  "entry": [{
    "fullUrl":
"http://test.fhir.ch/r4/Composition/2-1-ImmunizationAdministrationComposition",
    "resource": {
      "resourceType": "Composition",
      "id": "2-1-ImmunizationAdministrationComposition",
      "meta": {
```

## 2 · openEHR FLAT Composition (after openFHIR)

intermediate

```
{
  "ch_vacd_immunization_administration/medication
_management:0/vaccine_administered|code": "637",
  "ch_vacd_immunization_administration/medication
_management:0/vaccine_administered|terminology": "h
ttp://fhir.ch/ig/ch-vacd/CodeSystem/ch-vacd-swissme
dic-cs",
  "ch_vacd_immunization_administration/medication
_management:0/vaccine_administered|value": "Boostri
x",
  "ch_vacd_immunization_administration/medication
_management:0/time": "2017-09-15T07:12:56",
  "ch_vacd_immunization_administration/medication
_management:0/medication_details/lot_number/text_va
lue": "12-34244",
  "ch_vacd_immunization_administration/medication
_management:0/administration_details/route|code":
"20035000",
  "ch_vacd_immunization_administration/medication
_management:0/administration_details/route|terminol
ogy": "http://standardterms.edqm.eu",
  "ch_vacd_immunization_administration/medication
_management:0/administration_details/route|value":
"Intramuscular use",
  "ch_vacd_immunization_administration/medication
_management:0/dose_number": 1.
```

## 3 · EHRbase confirmation

AQL roundtrip

|                  |                                                            |
|------------------|------------------------------------------------------------|
| compositionUid   | a61d3e7d-ce8b-47e6-b969-7131db0ecalc::local.ehrbase.org::1 |
| ehrId            | f725dc67-394c-4f44-87f1-1617e13dd761                       |
| patientId (HAPI) | d1e3b8b8-5b27-4843-8648-a076f9634fd5                       |
| practitionerIds  | 6fe9d654-ca0b-47cc-a969-773015bcd95a                       |
| organizationIds  | a1a13537-8a68-4466-af49-304982e7a458                       |

### AQL roundtrip

```
{"meta":{"type":"RESULTSET","schema_version":"1.0.3","_created":"2026-05-17T13:47:50.673973879Z","_executed_aql":"SELECT c/uid/value AS uid, c/archetype_details/template_id/value AS template FROM EHR e[ehr_id/value='f725dc67-394c-4f44-87f1-1617e13dd761'] CONTAINS COMPOSITION c","resultsize":1},"q":"SELECT c/uid/value AS uid, c/archetype_details/template_id/value AS template FROM EHR e CONTAINS COMPOSITION c WHERE e/ehr_id/value = 'f725dc67-394c-4f44-87f1-1617e13dd761',"columns":[{"path":"c/uid/value","name":"uid"},{"path":"c/archetype_details/template_id/value","name":"templat
```

### How this works

- Client POSTs a CH VACD Immunization (or document Bundle) to `POST /Immunization`.
- PBLL fans Patient/Practitioner/Organization to `fhir-server-1` (HAPI). The FHIR server is the demographics + directory store.
- PBLL ensures an openEHR EHR exists in `EHRbase`, linked to the Patient via `EHR_STATUS.subject.external_ref`.
- PBLL calls `openFHIR /openfhir/toopenehr` with the Immunization to get a FLAT openEHR Composition.
- PBLL enriches FLAT with `ctx/feeder_audit/original_content` (verbatim FHIR JSON — Konkretisierung §13).
- PBLL persists the Composition to EHRbase and returns `201 Created` with `compositionUid`.

Hackathon harness — services: HAPI 8.8.1 · openFHIR 2.2.1 · EHRbase 2.31.0

screenshot: 02-01-immunization-administration-boostrix-done.png

# Comirnaty (COVID-19 mRNA) — CH VACD Bundle/document

Derived variant: Swissmedic 68225 + SNOMED 840539006 (COVID-19).

● CH VACD Pattern A — Platform Business Logic Layer (tracer bullet)

healthz metadata CH VACD IG ↗

Example: Immunization administration — comirnaty ▼

Convert &amp; Store ►

## 1 · FHIR Immunization (input)

application/fhir+json

```
{
  "resourceType": "Bundle",
  "id": "1-2-ImmunizationAdministration",
  "meta": {
    "lastUpdated": "2021-06-01T00:00:00.374+02:00",
    "profile": [
      "http://fhir.ch/ig/ch-vacd/StructureDefinition/ch-vacd-document-immunization-administration"
    ]
  },
  "identifier": {
    "system": "urn:ietf:rfc:3986",
    "value": "urn:uuid:5bca53b2-0973-4b91-961e-b1e93474c9c2-derived"
  },
  "type": "document",
  "timestamp": "2024-10-21T10:30:00+02:00",
  "entry": [
    {
      "fullUrl":
"http://test.fhir.ch/r4/Composition/2-2-ImmunizationAdministrationComposition",
      "resource": {
        "resourceType": "Composition",
```

## 2 · openEHR FLAT Composition (after openFHIR)

intermediate

```
{
  "ch_vacd_immunization_administration/medication
_management:0/vaccine_administered|code": "68225",
  "ch_vacd_immunization_administration/medication
_management:0/vaccine_administered|terminology": "h
ttp://fhir.ch/ig/ch-vacd/CodeSystem/ch-vacd-swissme
dic-cs",
  "ch_vacd_immunization_administration/medication
_management:0/vaccine_administered|value": "Comirna
ty (COVID-19 Vaccine, Pfizer)",
  "ch_vacd_immunization_administration/medication
_management:0/time": "2024-10-21T08:30:00",
  "ch_vacd_immunization_administration/medication
_management:0/medication_details/lot_number/text_va
lue": "HC9876",
  "ch_vacd_immunization_administration/medication
_management:0/administration_details/route|code":
"20035000",
  "ch_vacd_immunization_administration/medication
_management:0/administration_details/route|terminol
ogy": "http://standardterms.edqm.eu",
  "ch_vacd_immunization_administration/medication
_management:0/administration_details/route|value":
"Intramuscular use",
  "ch_vacd_immunization_administration/medication
_management:0/dose_number": 1.
```

## 3 · EHRbase confirmation

AQL roundup

|                  |                                                            |
|------------------|------------------------------------------------------------|
| compositionUid   | 4c71e232-75da-437b-85a6-33b0537b1bfb::local.ehrbase.org::1 |
| ehrId            | f3be0d60-b921-4599-9f62-3dc73b688358                       |
| patientId (HAPI) | 5d84fde8-3b3a-4f98-a0a6-a906077d60e0                       |
| practitionerIds  | cf82760a-0238-424e-bf37-232043cae639                       |
| organizationIds  | c7b835dd-ec5d-4826-9861-60ff23f2c208                       |

### AQL roundup

```
{"meta":{"type":"RESULTSET","schema version":"1.0.3","_created":"2026-05-17T13:47:50.994432963Z","_executed_aql":"SELECT c/uid/value AS uid, c/archetype_details/template_id/value AS template FROM EHR e[ehr_id/value='f3be0d60-b921-4599-9f62-3dc73b688358'] CONTAINS COMPOSITION c","resultsize":1},"q":"SELECT c/uid/value AS uid, c/archetype_details/template_id/value AS template FROM EHR e CONTAINS COMPOSITION c WHERE e/ehr_id/value = 'f3be0d60-b921-4599-9f62-3dc73b688358',"columns":[{"path":"c/uid/value","name":"uid"},"path":"c/archetype_details/template_id/value","name":"templat
```

### How this works

- Client POSTs a CH VACD Immunization (or document Bundle) to `POST /Immunization`.
- PBLL fans Patient/Practitioner/Organization to `fhir-server-1` (HAPI). The FHIR server is the demographics + directory store.
- PBLL ensures an openEHR EHR exists in `EHRbase`, linked to the Patient via `EHR_STATUS.subject.external_ref`.
- PBLL calls `openFHIR /openfhir/toopenehr` with the Immunization to get a FLAT openEHR Composition.
- PBLL enriches FLAT with `ctx/feeder_audit/original_content` (verbatim FHIR JSON — Konkretisierung §13).
- PBLL persists the Composition to EHRbase and returns `201 Created` with `compositionUid`.

Hackathon harness — services: HAPI 8.8.1 · openFHIR 2.2.1 · EHRbase 2.31.0

screenshot: 04-02-immunization-administration-comirnaty-done.png

# Priorix (MMR) — CH VACD Bundle/document

Derived variant: Swissmedic 615 + SNOMED 14189004/36989005/36653000.

● CH VACD Pattern A — Platform Business Logic Layer (tracer bullet)

healthz metadata CH VACD IG ↗

Example: Immunization administration — priorix ▼

Convert &amp; Store ▶

## 1 · FHIR Immunization (input)

application/fhir+json

```
{
  "resourceType": "Bundle",
  "id": "1-3-ImmunizationAdministration",
  "meta": {
    "lastUpdated": "2021-06-01T00:00:00.374+02:00",
    "profile": [
      "http://fhir.ch/ig/ch-vacd/StructureDefinition/ch-vacd-document-immunization-administration"
    ]
  },
  "identifier": {
    "system": "urn:ietf:rfc:3986",
    "value": "urn:uuid:5bca53b2-0973-4b91-961e-b1e93474c9c2-derived"
  },
  "type": "document",
  "timestamp": "2023-08-14T14:15:00+02:00",
  "entry": [
    {
      "fullUrl":
"http://test.fhir.ch/r4/Composition/2-3-ImmunizationAdministrationComposition",
      "resource": {
        "resourceType": "Composition",
```

## 2 · openEHR FLAT Composition (after openFHIR)

intermediate

```
{
  "ch_vacd_immunization_administration/medication
_management:0/vaccine_administered|code": "615",
  "ch_vacd_immunization_administration/medication
_management:0/vaccine_administered|terminology": "h
ttp://fhir.ch/ig/ch-vacd/CodeSystem/ch-vacd-swissme
dic-cs",
  "ch_vacd_immunization_administration/medication
_management:0/vaccine_administered|value": "Priori
x",
  "ch_vacd_immunization_administration/medication
_management:0/time": "2023-08-14T12:15:00",
  "ch_vacd_immunization_administration/medication
_management:0/medication_details/lot_number/text_va
lue": "MMR-2023-Q3",
  "ch_vacd_immunization_administration/medication
_management:0/administration_details/route|code":
"20035000",
  "ch_vacd_immunization_administration/medication
_management:0/administration_details/route|terminol
ogy": "http://standardterms.edqm.eu",
  "ch_vacd_immunization_administration/medication
_management:0/administration_details/route|value":
"Intramuscular use",
  "ch_vacd_immunization_administration/medication
_management:0/dose_number": 1.
```

## 3 · EHRbase confirmation

AQL roundtrip

|                  |                                                            |
|------------------|------------------------------------------------------------|
| compositionUid   | e5c93505-a0f6-4d36-8c0c-be4c81a01243::local.ehrbase.org::1 |
| ehrId            | 5365413b-8652-4d41-920c-b93d5c0c93ab                       |
| patientId (HAPI) | bd403ac9-56ed-4e6d-8776-e0037fbdac62                       |
| practitionerIds  | 711cbbe8-a825-4390-ac2e-12181cab913f                       |
| organizationIds  | c4509f82-2133-4189-b901-b60255f53571                       |

### AQL roundtrip

```
{
  "meta": {
    "type": "RESULTSET",
    "schema version": "1.0.3",
    "created": "2026-05-17T13:47:51.342195838Z",
    "executed_aql": "SELECT c/uid/value AS uid, c/archetype_details/template_id/value AS template FROM EHR e[ehr_id/value='5365413b-8652-4d41-920c-b93d5c0c93ab'] CONTAINS COMPOSITION c, \"resultsize\":1,\"q\":\"SELECT c/uid/value AS uid, c/archetype_details/template_id/value AS template FROM EHR e CONTAINS COMPOSITION c WHERE e/ehr_id/value = '5365413b-8652-4d41-920c-b93d5c0c93ab'\", \"columns\": [{\"path\":\"c/uid/value\", \"name\":\"uid\"}, {\"path\":\"c/archetype_details/template_id/value\", \"name\":\"templat
```

### How this works

- Client POSTs a CH VACD Immunization (or document Bundle) to `POST /Immunization`.
- PBLL fans Patient/Practitioner/Organization to `fhir-server-1` (HAPI). The FHIR server is the demographics + directory store.
- PBLL ensures an openEHR EHR exists in `EHRbase`, linked to the Patient via `EHR_STATUS.subject.external_ref`.
- PBLL calls `openFHIR /openfhir/toopenehr` with the Immunization to get a FLAT openEHR Composition.
- PBLL enriches FLAT with `ctx/feeder_audit/original_content` (verbatim FHIR JSON — Konkretisierung §13).
- PBLL persists the Composition to EHRbase and returns `201 Created` with `compositionUid`.

Hackathon harness — services: HAPI 8.8.1 · openFHIR 2.2.1 · EHRbase 2.31.0

screenshot: 06-03-immunization-administration-priorix-done.png

Built 2026-05-17 — Swiss GovTech Hackathon 2026 / DigiSanté

8 / 9



# What's Proven and What's Next

Acceptance criteria validated end-to-end against the live stack.

## Proven (all green)

- ✓ Stack comes up end-to-end on docker compose; PBLL fat JAR < 20 MB
- ✓ Bootstrap is idempotent (OPT + context.yml + model.yml on every restart, no duplicates)
- ✓ POST /Immunization returns 201 + compositionUid (warm path < 200 ms)
- ✓ feeder\_audit.original\_content contains the verbatim FHIR JSON (Konkretisierung §13)
- ✓ EHRbase composition queryable via AQL by ehr\_id
- ✓ Patient + Practitioner + Organization landed in HAPI from the document Bundle
- ✓ Strict mode: bare Immunization (non-document Bundle) rejected with 400 + OperationOutcome
- ✓ Demo UI drives all three canonical CH VACD document Bundles in Chromium
- ✓ Unit + integration tests pass against the live stack

## Deliberately out of scope (left to participating teams)

- Read path — GET /Immunization/{uid}, search by patient, Bundle re-hydration from fhir-server-1
- Profile validation — HAPI FhirValidator + CH VACD StructureDefinitions package
- Terminology lookups — live tx.fhir.ch/r4 calls for code system / value set checks
- Pattern B — inbound/outbound connectors and the sync-state table that goes with them